

An independent synthesis of publicly available observations and published analyses — not original telescope data. Figures are attributed to their sources, and established results are kept distinct from speculation.

Abstract

3I/ATLAS (C/2025 N1) is the third confirmed interstellar object to cross the Solar System, after 1I/'Oumuamua (2017) and 2I/Borisov (2019). Discovered by the ATLAS survey on 1 July 2025, it travels on an extreme hyperbolic orbit ($e \approx 6.1$) at a velocity at infinity near 58–60 km s^{-1} — by far the most dynamically extreme body yet recorded here. It passed perihelion on 29 October 2025 at $q \approx 1.36$ AU. JWST found a carbon-dioxide-dominated coma with a CO_2/H_2O ratio of 7.6 ± 0.3, among the highest measured in any comet. This report consolidates the public record (NASA/ESA, JPL Horizons, the Minor Planet Center, and 2025 JWST/Hubble results) and outlines a computational framework for trajectory and outgassing modelling — treating 3I/ATLAS as what the evidence supports, a natural interstellar comet, while addressing the artificial-origin speculation directly.

Key Orbital & Physical Parameters

Parameter	Symbol	Value	Source
Discovery	—	1 July 2025 (ATLAS, Chile)	Denneau+ 2025
Eccentricity	*e*	6.1–6.15 (strongly hyperbolic)	JPL / Seligman+
Perihelion distance	*q*	1.357 AU	JPL / MPC
Inclination	*i*	175.1° (retrograde)	JPL
Velocity at infinity	V_∞	≈ 58–60 km s^{-1}	Seligman+ 2025
Perihelion date	—	29 October 2025	JPL
Nucleus diameter	*D*	< 5.6 km (Hubble upper limit)	Jewitt+ 2025
Coma CO_2 / H_2O	—	7.6 ± 0.3 (among highest ever)	Cordiner+ 2025 (JWST)
Visible spectral slope	—	≈ 17 %/100 nm (red)	Seligman+ 2025

Composition & Origin

JWST NIRSpec (6 Aug 2025) resolved a CO_2-dominated coma containing water, CO, OCS, water ice and dust, with stronger outgassing on the sunward side. A CO_2/H_2O ratio of 7.6 ± 0.3 points to an intrinsically CO_2-rich nucleus — one that formed beyond the CO_2 frost line of a cold protoplanetary disk, or was processed by long radiation exposure. Some analyses tentatively link its kinematics to the Milky Way's older thick-disk population (Hopkins+ 2025), which would make it older than the Sun. Crucially, the bright, active coma — absent for 'Oumuamua — confirms ordinary cometary behaviour.

Is It Artificial?

The honest answer. Its speed and trajectory drew speculation that 3I/ATLAS might be technological — an echo of the 'Oumuamua debate led by Harvard's Avi Loeb. The scientific consensus is unambiguous: it is a natural comet. The decisive evidence is its own activity — a CO_2-driven coma, a dust tail, and sublimation chemistry no engineered object would mimic — and radio searches have found no technosignature. Even the preprint making the artificial case (Hibberd, Crowl & Loeb 2025) describes itself as *"largely a pedagogical exercise"* and concedes a natural comet is *"by far the most likely outcome."* The mature stance is to treat the artificial hypothesis as a low-probability prompt for sharper observation — not a co-equal explanation.

Computational Framework

Trajectory propagation. The object's motion is integrated as

$$a = -GM_\odot r/|r|^3 + \Sigma_{planets} + a_{NG} + a_{rel}$$

where a_{NG} is the non-gravitational (outgassing) term and a_{rel} a weak-field relativistic correction. An adaptive Runge–Kutta–Fehlberg (7/8) integrator conserves energy to $\Delta E < 10^{-9}$ J kg^{-1}, allowing post-perihelion reconstruction to ≤ 0.005 AU.

Relativistic note. Near-Sun light/path deflection from the Schwarzschild term is real but tiny at q ≈ 1.36 AU — far below the 1.75" solar-limb figure, and only marginally detectable by precision astrometry. **Capture test.** A 10^5-sample Monte-Carlo dispersion over (e, i, q, V∞) within observational uncertainty leaves ~99 % of realisations unbound — confirming no capture: 3I/ATLAS is leaving for good.

Implications

Three interstellar objects in under a decade implies the Solar System meets such bodies more often than classical capture models predicted (number density ~10^{-3} AU^{-3} for objects this size). Each is a free sample of another star system's chemistry, and their volatile and organic content keeps the panspermia question open — without implying any biological transfer. As survey power grows (notably the Vera C. Rubin Observatory), such detections should become routine, turning interstellar objects from rarities into a population studied statistically.

References

Seligman, Bolin et al., "Discovery and Preliminary Characterization of a Third Interstellar Object: 3I/ATLAS," arXiv:2507.02757 (2025). · Cordiner et al., "JWST Detection of a CO_2-Dominated Coma in 3I/ATLAS," arXiv:2508.18209 / ApJL (2025). · Jewitt et al., Hubble imaging & nucleus constraint (2025). · Hibberd, Crowl & Loeb, "Is the Interstellar Object 3I/ATLAS Alien Technology?" arXiv:2507.12213 (2025). · NASA / ESA 3I/ATLAS assessments (2025); JPL Horizons; Minor Planet Center Circulars (2025).